Manual

Graphic Analysis of Power Data

EMG-GPD 8.3

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Contents

1 Overview: Graphic Analysis of Performance Data

Overview

Purpose

This product contains graphic evaluations that show measured values over a period of time. You can separately select the data lines and combine them in diagrams for comparison.

Integration

For this product, the following requirements must be fulfilled:
- machine interface for data collection in the Process Communication Controller (PCC),
- license EMG-PDC to collect and process process data.

Features

This product provides efficient functions to analyze process values in detail with reference to other process values and logistical data from the production process.

- You can present the saved process values of a machine in a selectable period (line chart).
- You can display several diagrams in one report comparing the process values of a machine.

2 Graphic Process Analysis

Overview

Menu	Quality management → Process analysis → Graphic process analysis
Transaction code	pdc
Function authorization	pdc

The *Graphic process analysis* visually supports the data analysis.

Purpose

You can use this application to systematically survey processes using the measured values recorded for the different process characteristics. The objective of the process analysis is to identify interrelationships and properties of process characteristics and to get a better understanding of the process.

Integration

The application shows the measured values of process characteristics recorded via PCC.

Requirements

The measured values recorded are saved in the HYDRA database.

Selection criteria

The application provides the following selection criteria:

Tab *Machine*

Workplace

This selection criterion refers to the workplace where the measured values have been recorded.

Time domain

This selection criterion defines the evaluation period for which the measured values are displayed.

Compression value

This selection field is only available if the selection parameter *Automatic compression* is disabled. You configure the available compression values and their properties in the *Advanced object configuration*.

Automatic compression

If this option is enabled, the compression is automatically performed using the values of the other selection parameters. The selection field *Compression value* is disabled in this case.

Tab *Tag*

Tag type

This selection criterion refers to the tag type for which the measured values have been recorded. If a tag type is specified, then you must additionally specify the tag content as selection criterion.

Tag content

This selection criterion defines the value of the specified tag type for which the measured values are selected. You can also use a wildcard (*) as content.

If a value is specified for the tag content, then you must additionally specify the tag type as selection criterion.

Workplace

This selection criterion refers to the workplace where the measured values have been recorded.

Time domain

This selection criterion defines the evaluation period for which the measured values are displayed.

Compression value

This selection field is only available if the selection parameter *Automatic compression* is disabled. You configure the available compression values and their properties in the *Advanced object configuration*.

Automatic compression

If this option is enabled, the compression is automatically performed using the values of the other selection parameters. The selection field *Compression value* is disabled in this case.

Chart

Process parameter

This selection criterion specifies the process parameters. The measured values are then displayed for these process parameters. You can select up to 12 process parameters.

Legend

If this option is enabled, the legend (process parameter name and machine) is displayed in the chart.

Scale

You use this drop-down list to control the y-axis scaling of the graphic display. The following selection options are available:

- None
  If this option is selected, no y-axis scale is shown for any process data curve in the detail application.
- Consistent
  If this option is selected, one single y-axis scale is shown for all process data curves in the detail application.
- Individual
  If this option is selected, a different y-axis scale is shown for each process data curve.

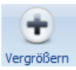
Toolbar

 **Add chart**

You use this button to add an additional data area of the detail application to separately show further process data curves.

 **Remove chart**

You use this button to remove the data area added last from the detail application.



Zoom in

Use this button to increase the time domain of the selection and to automatically request the data displayed.

Zoom out

Use this button to decrease the time domain of the selection and to automatically request the data displayed.

3 Histogram

Overview

Menu	Quality management → Process analysis → Histogram
Transaction code	pdh
Function authorization	pdh

Purpose

The *Histogram* report shows the distribution of measured values for a process parameter.

Integration

The histogram display is based on the data recorded for articles and machines.

Selection criteria

The application provides the following selection criteria:

Workplace

This selection criterion refers to the workplace where the measured values have been recorded.

Process parameter

This selection criterion specifies the process parameter. The measured values are then displayed for this process parameter.

Time from,to

This selection criterion defines the evaluation period for which the measured values are displayed.

Toolbar

Histogram settings

Use this button to call the settings of the histogram display.

Detail application *Histogram*

The histogram is always based on the total of available measured values matching the selection criteria. The number of classes and the additionally displayed information influence the display of the histogram. Call the settings dialog of the "histograms" to define the contents of this application. If you use this dialog to make changes, the changes are saved per user.

Detail application *Histogram settings*

Field descriptions

Number of classes

Specifies the number of histogram classes. The measured values are classified and displayed in these classes.

Show histogram title

If this option is enabled, the histogram title is shown on top of the histogram display.

Histogram title

This entry specifies the histogram title displayed. This entry is only relevant if the option *Show histogram title* is enabled.

Consider number of decimal places

If this option is enabled, the specified number of decimal places is used to identify the histogram classes. If this option is disabled, integer values are used to identify and display the classes.

Number of decimal places

Specifies the number of decimal places that are used. This entry is only relevant if the option *Consider number of decimal places* is enabled.

Show bars

This option controls if the histogram shows the values of the classes in a bar chart.

Show envelope

If this option is enabled, an envelope is plotted over the values of the classes.

Show grid

Use this option to show or hide a grid in the background of the histogram.

Y-axis labeling

Controls the Y-axis values of the histogram. The following values can be set:

- None
- Relative frequency

- Frequency in %

X-axis labeling

Controls the X-axis values of the histogram. The following values can be set:

- None
- Class limits